

Bengaluru Infrastructure Accountability Tracker — Build Spec

2026-09-19 · @Someone

1. Purpose and scope

A public, free website showing what Karnataka's agencies have committed to building in Bengaluru, what each project cost, how far behind it is, and what is still pending — so that residents, journalists and civic groups can ask specific, evidenced questions instead of general ones.

Who it is for, in priority order

1. **Residents** who want the status of a project near them, and language they can take to a corporator or MLA.
2. **Journalists and researchers** who need a sourced, exportable dataset rather than a story.
3. **Resident welfare associations and civic groups** who already do this work and currently rebuild the same data each time.

Scope

- Geography: the Greater Bengaluru Authority area plus the BDA peripheral area.
- Agencies: BMRCL, BMTC, BDA, BWSSB, K-RIDE, B-SMILE, GBA and the five city corporations (and BBMP historically).
- Categories: metro and rail, roads and corridors, buses, water and sewerage, stormwater drains and lakes, parks and public space.
- Time: FY2020-21 forward at launch, extended backwards as older documents are located.

Explicitly out of scope

- Allegations of corruption, or any claim about *why* a delay happened.
- Contractor performance rankings.
- Anything sourced only from social media or unattributed claims.

What success looks like

- A resident finds their ward's projects in under a minute.
- A journalist cites the site, or exports the dataset, without re-verifying it.
- An agency engages with a correction rather than ignoring the site.

- The quarterly update ships on time, four times running.

2. Editorial principles

These are not style preferences. They determine the schema and the UI, and they are what makes the site defensible if an agency disputes it.

P1 — Every number carries its source, its as-of date, and its type. Never a bare figure. Announced, sanctioned, released and utilised are four different things and are never merged into one number or one chart line.

P2 — Allocation is not spend. An allocation is what was announced in a budget. It is not evidence that money was spent. The site says so wherever a figure appears, and the absence of utilisation data is itself displayed — "released: no data" is a finding, not a blank.

P3 — Definitions change between years; never silently join them. The "Bengaluru grant" meant different things in FY2021-22 and FY2025-26. Any time series that crosses a definition change must break the line or flag it. Where figures are not comparable, the site says they are not comparable.

P4 — Confidence is visible. Each record carries a confidence level: *high* (official document), *medium* (reported in credible media), *low* (announcement only, unverified). This is shown in the UI, not buried in metadata.

P5 — State facts, never infer intent. "Lowest bid was ₹22,267 crore against an estimate of ₹17,698 crore" is publishable. Any characterisation of why is not.

P6 — Attribute reported claims. "Deccan Herald reported X" carries different weight from "X is true", and the site preserves the difference.

P7 — Stale is worse than absent. Every project shows its own *verified on date*. A record not re-verified in the last two cycles is visually marked as possibly outdated.

P8 — Corrections are public. A logged, dated correction is the strongest defence available and the clearest signal of good faith.

P9 — Show the full history, across governments. Delays span administrations. Pre-2020 data is not a nice-to-have; it is the main defence against the site being read as partisan.

3. Data model

Three related entities: **project**, **money_entry**, **event**. Money and events are separate records rather than fields on the project, so that history is preserved and every figure can carry its own source and date.

3.1 project

Field	Type	Notes
id	slug	Stable, human-readable: metro-phase-2
name	string	As the agency calls it
aka	string[]	Alternate names for search
agency	enum	BMRCL, BMTCL, BDA, BWSSB, K-RIDE, B-SMILE, GBA, corporation, other
category	enum	metro_rail, roads, buses, water_sewerage, drains_lakes, parks_public
scope	string	"75.09 km, 4 extensions, 2 new lines"
status	enum	announced, sanctioned, tendering, awarded, under_construction, partially_open, complete, stalled, cancelled
sanction_date	date	Nullable
original_deadline	date	Nullable
current_deadline	date	Nullable
slippage_months	derived	current_deadline – original_deadline
pending	string	What is blocking it, factually stated
geometry	GeoJSON	LineString for corridors, Point for facilities, Polygon for areas
wards	string[]	Ward codes the project touches
corporation	enum	Central, East, North, South, West, outside_GBA
confidence	enum	high, medium, low
verified_on	date	Last date a human checked this record
notes	string	Caveats, conflicting figures

3.2 money_entry

One row per figure. A project has many.

Field	Type	Notes
<code>project_id</code>	slug	FK
<code>fiscal_year</code>	string	2025-26
<code>kind</code>	enum	announced, sanctioned, revised_cost, bid_received, released, utilised
<code>amount_cr</code>	number	₹ crore
<code>funder</code>	enum	state, centre, agency_borrowing, external (JICA, World Bank, HUDCO), private
<code>source_id</code>	string	FK to source
<code>as_of</code>	date	When the figure was true
<code>comparable_to_prior</code>	boolean	False where the definition changed (P3)

3.3 event

One row per dated development. Drives the timeline and the "what changed" feed.

Field	Type	Notes
<code>project_id</code>	slug	FK
<code>date</code>	date	
<code>type</code>	enum	sanctioned, tender_floated, bid_received, awarded, deadline_revised, section_opened, cost_revised, stalled
<code>summary</code>	string	Factual, one sentence, no characterisation
<code>source_id</code>	string	FK

3.4 source

Field	Type	Notes
<code>id</code>	string	S1, S2 ...
<code>title</code>	string	
<code>publisher</code>	string	

Field	Type	Notes
doc_type	enum	budget_document, audit_report, agency_report, assembly_reply, rti_response, news, press_release
url	string	
page_ref	string	Page or section, for PDFs
published_on	date	
retrieved_on	date	
archive_url	string	Wayback snapshot — essential, government URLs rot

doc_type drives confidence : budget_document, audit_report, assembly_reply and rti_response → high; agency_report → high or medium; news → medium; press_release alone → low.

4. Data sources

4.1 Priority order

1. **Karnataka Finance Department budget documents**, including the annual Link Documents with scheme-wise allocation. Go back well before 2020. Best source for announced and sanctioned .
2. **CAG audit reports on Karnataka**. The only source that reliably compares sanctioned against spent, and often already contains findings on delay. Best and usually only source for utilised .
3. **OpenCity.in**, which hosts BBMP budgets from 2016-17 onward. Treat as a potential collaborator, not a scrape target — approach them before launch.
4. **Agency annual reports** — BMRCL, BWSSB, BMTC, BDA. Good for physical progress and fleet numbers.
5. **Assembly question replies**. The single best source for project-wise status, and usually published as PDFs.
6. **RTI responses**, for the gaps nothing else fills. Slow, but this is what makes the site original rather than a news aggregator.
7. **Credible news reporting**, for events and current status between official releases. Always medium confidence.

4.2 Realistic expectations

- Pre-2020 coverage will be good for metro and water, thin for roads, and poor for anything at ward level.
- `utilised` figures will be missing for most projects most of the time. That absence is displayed, not hidden (P2).
- Some government sites serve PDFs poorly to automated fetchers, and some are behind session tokens. Assume a share of collection is manual.
- Government URLs rot. Every source gets a Wayback snapshot at collection time.

4.3 Starting dataset

A working spreadsheet already exists covering FY2020-21 to FY2026-27: roughly 22 allocation rows and 18 projects, each with a source ID and URL, and an explicit gaps list. It is entirely `medium` confidence (news-derived) and should be treated as a skeleton to be replaced field by field with `high` confidence figures, not as the launch dataset.

4.4 Collection workflow

For each project: locate the primary document → extract figures with page references → snapshot the URL → record `as_of` and `retrieved_on` → set confidence from `doc_type` → flag any conflict with an existing figure in `notes` rather than overwriting it.

5. Site modules

5.1 Map

- Layers toggled by agency; colour by **status**, not agency, since status is what people came for.
- Hover: name, latest cost, status. Click: project card with the money summary and a link to the full page.
- **Ward filter** — the single highest-value interaction, because it turns a city dashboard into "my area". Drives repeat visits.
- Confidence shown on the marker (e.g. dashed outline for `low`).
- Corridors as lines, facilities as points. Where alignment is unknown, show an approximate point and say so rather than drawing a false line.

5.2 Table

- Same records, sortable and filterable, with CSV and JSON export.

- Will do more real work than the map for journalists and RTI activists. Sorting by `slippage_months` is, by itself, a story.
- Columns: project, agency, sanctioned, latest cost, overrun %, status, slippage, verified on, confidence.

5.3 Project page

One stable URL per project. This is the page people share and cite, so it matters more than the map.

- Money timeline: announced → sanctioned → revised → bid → released → utilised, each with source and date.
- Deadline history, showing every revision.
- What is pending, stated factually.
- Full source list with archive links.
- "Last verified" and a link to the corrections log.

5.4 What changed feed

Dated `event` records across all projects, filterable by agency. Lower priority than the first three — consider launching without it and adding once the base is stable.

5.5 Ask your representative

Generates a factual summary of a project plus a template letter or RTI application.

Constraint: the generated text must be strictly factual and neutral. A template that alleges wrongdoing puts your words into hundreds of letters over your name. It should ask for information and status, not make accusations.

5.6 Cross-cutting

- Kannada alongside English. A civic accountability site for Bengaluru that is English-only excludes much of the people it claims to serve. At minimum: interface, disclaimer and project status labels.
- Mobile first. Most users will arrive on a phone from a WhatsApp link.
- Every page states the site's update cadence and its own last-verified date.

6. Update process

6.1 Cadence

Quarterly, aligned to when the data actually moves: the Karnataka budget lands in March, agency annual reports follow, and Assembly sessions produce question replies in batches.

Suggested windows: **April** (post-budget, the big one), **July, October, January**.

6.2 Separate the clock from the content

Some fields move quarterly (allocations, utilisation); others move weekly (tender awarded, cabinet decision, line opened). Publishing everything quarterly means that when a metro line opens in month two, the site is visibly wrong and people notice fast.

So:

- **Per-project** `verified_on`, not one site-wide date.
- **Out-of-cycle corrections are always allowed** for factual changes. The quarterly cycle governs the systematic re-verification sweep, not the ability to fix something.
- A record not touched for two cycles is flagged in the UI as possibly outdated.

6.3 Publish the cadence as a promise

A visible "updated quarterly · next update due [date]" is itself a credibility signal, and it inverts the usual complaint about government dashboards. Miss it twice and the project is dead, so quarterly is the right call for something run alongside other work.

6.4 The quarterly sweep

1. Pull new budget and audit documents; add `money_entry` rows.
2. Re-verify status on every project; update `verified_on` even where nothing changed.
3. Add `event` rows for the quarter's developments.
4. Re-snapshot any source URL that has moved or died.
5. Publish a short changelog naming what changed and what could not be verified.

6.5 Corrections log

Public, dated, permanent. Each entry: what was wrong, what it now says, who raised it, what source resolved it. Prompt documented correction is the strongest defence available and the clearest signal of good faith.

7. Legal posture and disclaimer

A lawyer reviews this before launch. The draft below is a starting point for that conversation, not a substitute for it.

7.1 Questions to put to counsel

Beyond a general review, three specific issues that a broad "is this okay" review may skip:

1. **Naming private contractors.** Agencies and officials acting in a public capacity are one risk profile; a private company whose bid the site characterises is another.
2. **User-generated content.** If comments or citizen reports are ever added, intermediary liability under the IT Act changes the obligations considerably. Worth deciding now, since it affects architecture.
3. **Republishing government documents.** Mostly fine, but confirm what can be hosted directly versus linked.

7.2 What protects the site, in practice

More than the disclaimer wording:

- **Truth is a complete defence** in Indian defamation law — so sourcing discipline is the real protection.
- **A visible corrections log** and prompt response to disputes.
- **No inference about intent** (P5). State the gap; never characterise why.
- **Attribution of reported claims** (P6).
- **Care with named individuals.** Projects and agencies are safe ground. Naming officials or contractors in a way that implies blame is where risk concentrates.

7.3 Draft public disclaimer

About this data

This site compiles publicly available information about infrastructure projects in Bengaluru from government budget documents, agency reports, audit reports and published news coverage. Every figure is shown with its source and the date it was recorded.

Figures are drawn from official documents and media reports as published. They are not independently audited by us. Budget allocations, sanctioned amounts, funds released and actual expenditure are different things, and we label which is which. An allocation is not evidence that money was spent, and a delay is not evidence of wrongdoing.

Project timelines change. Information is accurate as of the "verified on" date shown for each project and may be outdated. We update this site quarterly.

We do not allege misconduct by any individual, official, department or contractor. Infrastructure delays have many causes, including land acquisition, litigation, clearances, funding, contractor capacity and design changes.

If you are an agency or individual and believe something here is inaccurate, write to [email]. We will review it against our sources and publish a correction where warranted. Corrections are logged publicly.

Place it in the footer of every page, on an "About the data" page, and in abbreviated form on each project page near the figures.

8. Technical architecture

8.1 Principle

The site should hold itself to the standard it asks of the government. Data in Git, every change auditable, full history public. That property is the argument.

8.2 Shape

- **Data as flat files in a public Git repo** — JSON or CSV per entity, plus GeoJSON for geometry. Human-editable, diffable, forkable, and citable at a commit.
- **Static site generated from that data.** No runtime database, no server to maintain, near-zero hosting cost, and it survives traffic spikes when a story links to it.

- **Map:** MapLibre GL with an open basemap; avoid a vendor with usage-based pricing that can be weaponised by traffic.
- **Hosting:** static, on infrastructure already in use.
- **Open the data explicitly.** A clear licence (CC BY or ODbL) invites reuse. Reuse is how a project like this outlives its founder's attention.

8.3 Ward and geometry data

Ward boundaries changed with the GBA restructuring, so keep both the pre-GBA BBMP wards and the new five-corporation divisions. A project spanning years needs both to be findable. OpenCity and the Karnataka GIS portal are the likely sources.

8.4 Build and validation

A build-time validator that fails the build on: a `money_entry` with no `source_id`; a `source` with no `retrieved_on`; a project with no `verified_on`; a `confidence` value inconsistent with its `doc_type`. Enforcing P1 mechanically is more reliable than enforcing it by discipline.

9. Build phases

Phase 1 — Metro only. Table and project pages. No map.

BMRCCL data is the best sourced and carries the most public interest, and a table plus project pages proves the format at a fraction of the effort of a map. Deliverables: schema implemented, validator running, 8–12 metro projects at `high` confidence, disclaimer and corrections log live, quarterly cadence published.

The temptation will be to build the map first because it demos well. Resist it: the map is worth little over thin data, and thin data on a map looks authoritative in a way that thin data in a table does not.

Phase 2 — Add the map and the remaining agencies.

Geometry for metro lines and major corridors. BMTC, BDA, BWSSB, K-RIDE, B-SMILE records. Agency layer toggles and status colouring.

Phase 3 — Ward filtering and the representative tools.

Ward boundary data, ward-scoped views, generated summaries and letter or RTI templates. This is the phase that converts browsing into action.

Phase 4 — Event feed and pre-2020 history.

The "what changed" timeline and backwards extension of the record. Both are ongoing work rather than a finite deliverable.

Before Phase 1 ships: legal review, and a decision on Kannada — retrofitting localisation is far more expensive than building for it.

10. Open questions and risks

10.1 To decide before building

- **Kannada at launch, or English first?** Retrofitting localisation is expensive. This affects Phase 1.
- **Named individuals: in or out?** Recommendation is out. Projects and agencies only.
- **Who is the publisher?** A named individual, Redmonk, or a separate civic entity. This affects both legal exposure and how the site is perceived politically.
- **Approach OpenCity as a collaborator?** They hold data the site needs and have been doing this longer. Partnering is better than parallel effort.
- **Comments or citizen reports?** Recommendation is no, at least initially — it changes the legal position materially.

10.2 Standing risks

Maintenance is the real cost, not the build. A stale civic dashboard is worse than none, because it becomes a source of wrong numbers that people cite. The quarterly commitment is the hardest part of this project and the one most likely to fail.

Perceived partisanship. Delays span multiple governments. Pre-2020 history is the main defence, which is another reason to prioritise it over new features.

Being wrong in public. A citizen taking a wrong number to an MLA damages the project permanently. This is what P1, P4 and the validator exist to prevent.

Data source fragility. Government URLs rot and portals change. Archive snapshots at collection time are not optional.

Agency response. Most likely outcome is being ignored. Second most likely is a dispute over a specific figure — which is why the corrections process must exist before launch, not after the first complaint.